

Waking Up Twice: A Test Article for the JMP Quarto Format

ANONYMOUS SUBMISSION

ABSTRACT

This is a test article for the Quarto wrapper around the *Journal of Mathematical Philosophy* LaTeX class. It exercises theorems, a restated proposition in an appendix, equations, a figure, a table, citations, and back matter. It says nothing about Sleeping Beauty that has not been said.

1. Introduction

The Sleeping Beauty problem has been with us for a while (Haugerud 2001, 34). Renner-Alvi (2023, 119) thinks it is still open, and so do others (see Batzorig 2019, ch. 2; Tanaka 2025). Section 2 sets out a model, and section 3 states the main results. The longer proofs are in appendix A.¹

Here is a quotation long enough to be set as a block:

[The thirder] is committed to the claim that an agent can lower her credence in a proposition without acquiring any evidence against it, merely by coming to occupy one of several centered worlds compatible with what she already believed. ... Whether that is a reductio or a discovery is precisely what is at issue. (Renner-Alvi 2023, 119)

And an inline one: “a case in which learning nothing is not the same as learning that nothing has happened” (Haugerud 2001, 34).

¹ A footnote, placed after the punctuation, as the guide asks.

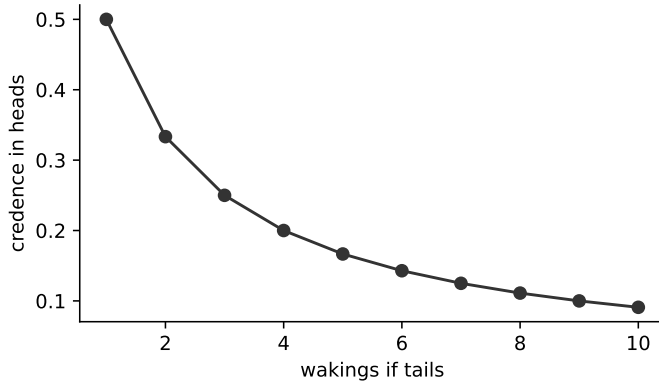


Figure 1: Credence in heads under (1), by the number of wakings if the coin lands tails.

2. The model

Definition 1 (Vigil). A *Vigil* is a triple $\langle W, \text{Cr}, \tau \rangle$ in which W is a finite set of centered worlds, Cr a probability measure on $\mathcal{P}(W)$, and $\tau : W \rightarrow \mathbb{N}$ assigns to each world the number of times the agent is woken in it.

The update rule we care about is

$$\text{Cr}^+(A) = \frac{\sum_{w \in A} \tau(w) \text{Cr}(\{w\})}{\sum_{w \in W} \tau(w) \text{Cr}(\{w\})}, \quad (1)$$

where Cr^+ is the credence on waking. Figure 1 plots what (1) says about heads as the number of tails wakings grows.

3. Results

Lemma 2. *Each waking in a Vigil in which τ is constant on T has the same weight.*

Proposition 3 (Thirder). *Let $\langle W, \text{Cr}, \tau \rangle$ be a Vigil in which τ takes exactly the values 1 and 2. If credence on waking is proportional to τ , credence in heads is $1/3$.*

Proof. By Lemma 2 the three wakings are equiprobable, and exactly one of them is in H . □

Theorem 4. *Let $\langle W, \text{Cr}, \tau \rangle$ be a Vigil with $\tau(w) \leq k$ for every $w \in W$. Then iterated proportional updating converges, and its limit is independent of Cr .*

Theorem 4 is proved in appendix A. Table 1 compares the rules, and Proposition 3 is the familiar thirder verdict.

Rule	Prior	On waking	After learning it is Monday
Halfer	1/2	1/2	2/3
Thirder	1/2	1/3	1/2
Double halfer	1/2	1/2	1/2

Table 1: Credence in heads under three rules.

4. Conclusion

That is all.

A. Proofs

Theorem 4. *Let $\langle W, Cr, \tau \rangle$ be a Vigil with $\tau(w) \leq k$ for every $w \in W$. Then iterated proportional updating converges, and its limit is independent of Cr .*

Proof. Since τ is bounded by k and W is finite, the map $Cr \mapsto Cr^+$ defined by (1) is a continuous self-map of the simplex, and the rest is Banach. $\square$

References

- Batzorig, Naranbaatar. 2019. *Coherence without Certainty*. 2nd ed. Kerneberg University Press. <https://doi.org/10.00000/kup.2019.0114>.
- Haugerud, Ingvid. 2001. “Self-Location and the Vigil.” *Inquiries in Formal Epistemology* 12 (1): 33–41. <https://doi.org/10.00000/ife.2001.12.33>.
- Renner-Alvi, Dorothea. 2023. “Non-Ideal Agents and the Demands of Coherence.” *Ratio Formalis* 58 (3): 214. <https://doi.org/10.00000/rf.2023.58.214>.
- Tanaka, Miho. 2025. “A Topology for Self-Locating Credence.” Preprint, arXiv, March 3, 2025. <https://doi.org/10.00000/arXiv.2503.01947>.